Data Mining and Discovery – GANGIREDDY NAGARJUNA REDDY (20544955)

Immigrant Support Referral System

Introduction: In this project I created Immigrant Support Referral System ISRS a SQLite database to manage support service referrals for immigrants in the UK. This project aims to find a balance between administrative efficiency and data ethics so that vulnerable populations can safely access health care, legal and educational support without worrying about their privacy.

Database Schema: in this created a database with normalized schema of five tables,the Basic demographic data is stored safely in clients table with an auto incrementing client_id as primary key, the Organizations are kept in service_providers lookup table.

The referrals table is the primary table of database. There are nearly individual count of 1200 records in the referrals table which meets the assignment requirements. The table acts as a transaction hub and links both client and specific service provider through foreign keys. To handle these situations where a single referral involves multiple categories of support, I created a referral_tags table instead of duplicating referral records. this table uses composite primary key consisting referral_id, tag to connect multiple attributes in clean way. Finally, the raw_referral_import table is the staging area for data intake.

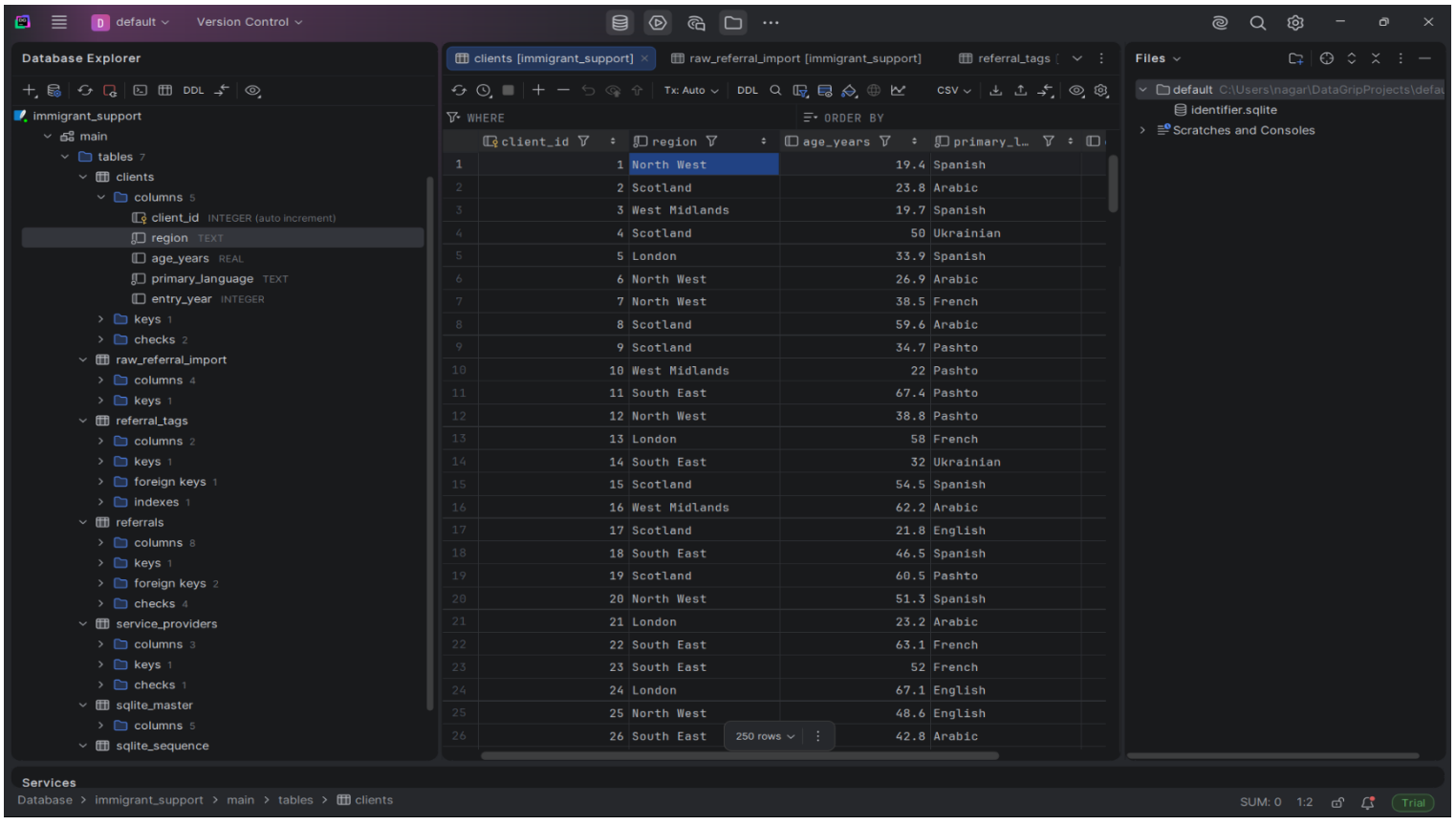


Figure 1: creation of table.

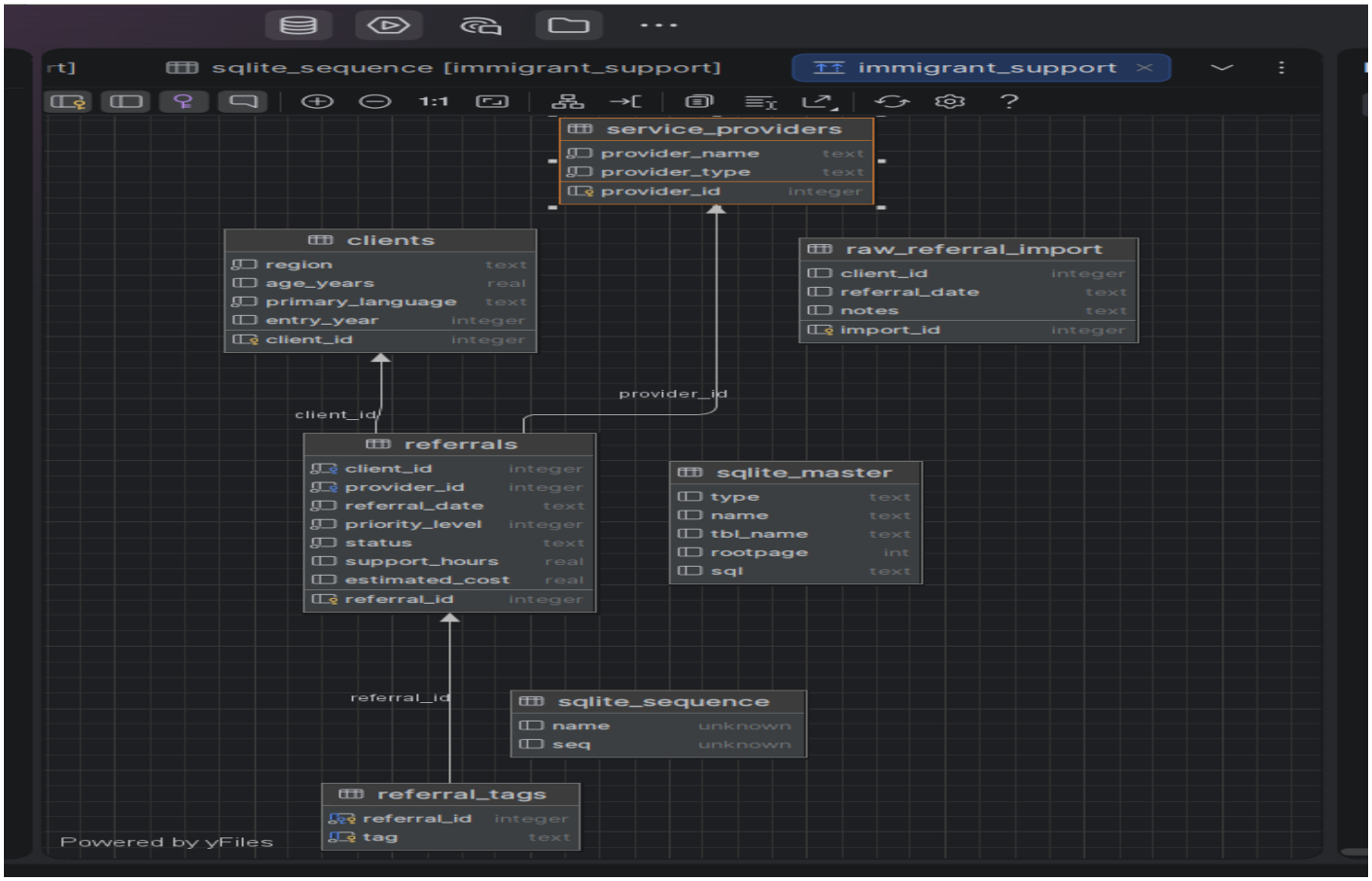


Figure 2: Entity Relationship Diagram of the main tables, foreign keys and relational links..

The database is highly normalized in 5 interrelated tables to ensure strict referential integrity and no data redundancy. Separating entities into separate tables is good for database scalability and helps prevent update anomalies..

- **clients:** Demographic data is anonymised. This is separate from the referrals table so that a client's core details are only stored once if they need to be referred multiple times.
- **service_providers:** a lookup table by type of aid the organization provides e.g. Legal, Healthcare, which allows to standardize on naming conventions and easily update provider information.
- **referral:** This is the main table that connects clients and services which has 1200 records which is a lot more than the 1000 rows that are required and The clients and service providers are connected by foreign keys.
- **referral_tag:** this table is used to solve many to many relationship where You can add multiple tags to a single referral without creating a duplicate referral record..
- **raw_referral_import:** Initial staging area for the data which is brought in before cleaning.

Data Types: database includes all four data measurement levels :

- Nominal: Categorical data without any natural order, e.g. region, provider_type.
- Ordinal: The column priority_level gives the case urgency level from 1 to 5.
- Interval: this referral_date field measures time frames difference between dates has mathematical significance but there is no absolute zero..

- Ratio: Quantitative variables like age_years and support_hours have a true zero point so you can compare them meaningfully using ratios like you can compute double the support hours.

Data types	Columns	Explanation
nominal	Region	Nominal qualitative data that has no numerical order or ranking e.g. 'London', 'Legal'.
ordinal	Priority_level	The Ranked qualitative data where order matters Scaled 1 to 5 where 5 being the most urgent. The exact distance between ranks is not defined.
interval	Referral_date	Quantitative data where the distance between two dates is meaningful but there is no true zero point.
ratio	Age_years, support_hours	Quantitative data with an absolute zero point so that ratios can be meaningful (e.g. 10 hours is exactly twice as long as 5 hours)

Data Generation: the data used in this project was ddeveloped synthetically to maintain privacy. But in the real world administrative data is perfect so I deliberately incorporated realism into the data set. I simulated incomplete client intake forms by leaving 5% of age_year column and 10% of the support_hour column to blank. missing values has been added to null into the database and also built a staging table raw_referral_import with a 20 sets of duplicate records. This was a planned choice to demonstrate how SQL duplication methods such as GROUP BY or ROW_NUMBER() can be used to clean up or remove all the messy data before it enters the main system.

Conclusion: The ISRS database proves that a strong approach to database management is possible through a highly normalized schema with strict referential integrity which guarantees data consistency and reliability. finally introducing data imperfections such as missing entries, duplicates etc to show the importance of proper data cleaning methods in real world settings.